What about video memory?

To run Stable Diffusion, the amount of VRAM (Video RAM) you need depends on the size of the images you want to generate. A minimum of 8 gigabytes of VRAM is recommended, but for producing larger 512x512 images without adjusting settings, a GPU with 12 gigabytes of VRAM or more is required. The RTX 3060 with 12 gigs of VRAM is a budget-friendly option, but it may sacrifice speed and gaming performance compared to the RTX 3060 Ti or 3080 Ti. If you want to generate larger images and have a bigger budget, the RTX 4090 with as much VRAM as possible would be an ideal choice. Although some optimized forks require less VRAM, it is recommended to stick with RTX NVIDIA cards with at least 8 gigabytes of memory for compatibility.

Should You Use an Optimized Fork of Stable Diffusion?

Simply put, yes. The Stable Diffusion community has made great strides in expanding GPU support and accessibility. Community forks offer additional features like a user interface, more models, and optimizations that allow users to generate larger images with less VRAM. Some users have been able to generate 512×512 images with as little as 4 gigabytes of VRAM using community forks. The user interface also makes using Stable Diffusion and generating AI video possible. However, it's important to be cautious as some modifications may be malicious. Antivirus warnings should not be ignored. With the right hardware, you can focus on optimizing your Stable Diffusion prompts rather than your PC.

Sample Script 1

- `import torch`
- `import torch.nn as nn`
- `import torch.optim as optim`
- `import torchvision.transforms as transforms`
- `from torchvision.utils import save _ image`
- `from torch.utils.data import DataLoader`
- `from tqdm import tqdm`